

SHETH L.U.J. AND SIR M.V. COLLEGE

Aim : Practical 3: Simulation of Single Mote using TOSSIM

Code:

```
print("Yukta Sonawane T122")
import heapq

class MockNode:
    """Stands in for a single TOSSIM node (e.g. node 0 running BlinkC)."""

    def __init__(self, node_id):
        self.node_id = node_id
        self.boot_time = None

    def boot_at_time(self, time):
        self.boot_time = time

class MockTossim:
    """
    Stands in for the Tossim object. Holds a priority queue ("event
    queue")
    of (time, description) pairs -- this mirrors how real TOSSIM jumps
    directly from event to event rather than advancing time continuously.
    """

    TIMER_PERIOD = 512

    def __init__(self):
        self.event_queue = []
        self.nodes = {}

    def get_node(self, node_id):
        node = MockNode(node_id)
        self.nodes[node_id] = node
        return node

    def schedule_boot(self, node):
        heapq.heappush(
            self.event_queue,
            (
```

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```
        node.boot_time,
        f"Booting node {node.node_id} at time {node.boot_time}."
    )
)

first_fire = node.boot_time + self.TIMER_PERIOD

heapq.heappush(
    self.event_queue,
    (
        first_fire,
        f"BlinkC: Timer 0 fired @ {first_fire}."
    )
)

def run_next_event(self):
    """Pop and print the earliest event, then schedule the next timer
    fire."""
    if not self.event_queue:
        return False

    time, description = heapq.heappop(self.event_queue)
    print(f"DEBUG ({self._active_node_id()}) : {description}")

    if "Timer 0 fired" in description:
        next_fire = time + self.TIMER_PERIOD

        heapq.heappush(
            self.event_queue,
            (
                next_fire,
                f"BlinkC: Timer 0 fired @ {next_fire}."
            )
        )

    return True

def _active_node_id(self):
```

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```
        return next(iter(self.nodes), 0)

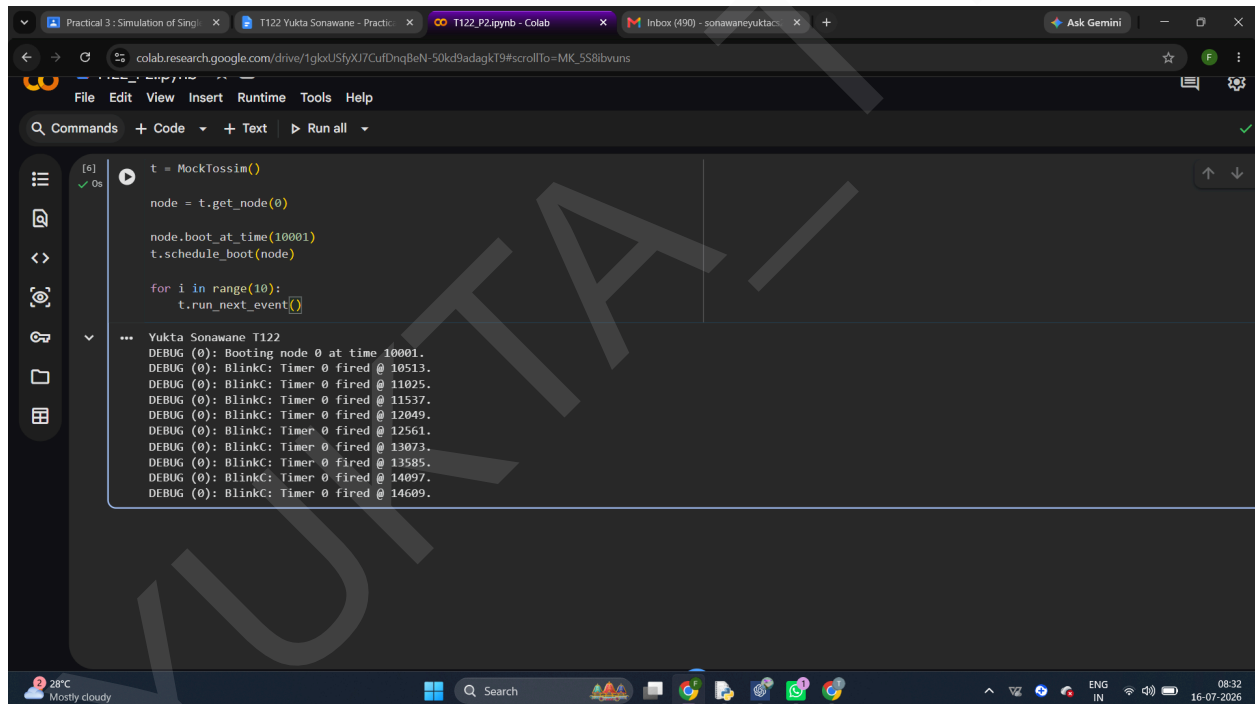
t = MockTossim()

node = t.get_node(0)

node.boot_at_time(10001)
t.schedule_boot(node)

for i in range(10):
    t.run_next_event()
```

Output:



```
[0] t = MockTossim()
    node = t.get_node(0)
    node.boot_at_time(10001)
    t.schedule_boot(node)

    for i in range(10):
        t.run_next_event()

... Yukta Sonawane T122
DEBUG (0): Booting node 0 at time 10001.
DEBUG (0): BlinkC: Timer 0 fired @ 10513.
DEBUG (0): BlinkC: Timer 0 fired @ 11025.
DEBUG (0): BlinkC: Timer 0 fired @ 11537.
DEBUG (0): BlinkC: Timer 0 fired @ 12049.
DEBUG (0): BlinkC: Timer 0 fired @ 12561.
DEBUG (0): BlinkC: Timer 0 fired @ 13073.
DEBUG (0): BlinkC: Timer 0 fired @ 13585.
DEBUG (0): BlinkC: Timer 0 fired @ 14097.
DEBUG (0): BlinkC: Timer 0 fired @ 14609.
```